
Design and Simulation Walkthrough: Superconducting Quantum Devices in Keysight ADS QuantumPro

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August 15, 2026

ABSTRACT

TODO: 3–5 sentences summarizing the device you design, the QuantumPro workflow you follow, and the main simulation results.

Keywords superconducting qubits · QuantumPro · ADS · transmon · CPW resonators · EPR · flip-chip

Introduction

- Motivation: why this device / process matter
- Brief circuit-QED context (qubit + readout resonator)
- Software stack: PathWave ADS + QuantumPro
- Design cycle overview:
 1. Schematic / Quantum Artwork layout
 2. Technology stackup + ports
 3. Linear EM (JJ $\rightarrow$ abstract L_J)
 4. Extract S-parameters / quantum parameters
 5. Iterate geometry; export GDS
- Scope of this walkthrough (what you will build and simulate)
- Roadmap of remaining sections

Device Targets

- Chip / process assumptions (substrate, metal, temperature)
- Target qubit frequency, anharmonicity α , T_1 goals
- Target resonator frequency, κ , dispersive shift χ
- Coupling targets (g , nearest-neighbor if multi-qubit)
- Topology choice (planar single-layer vs flip-chip; Xmon vs pocket; etc.)

Layout Design

Defining Layers

- Create workspace; select Quantum Technology Library (Single-Layer / Multi-Layer)
- Open `tech:subst`: thicknesses, ε_r , loss tangent, PEC conductors
- Layer roles to document (fill table or bullets as you go):
 1. boundary
 2. conductor / cond
 3. keepout
 4. ground fill
 5. airbridge
 6. inductor (JJ placeholder)
 7. air / dielectric
- Choose CPW W , g for $50 \sim \Omega$; note Z_0 , ε_{eff} , L' , C'
- Figure / screenshot: stackup editor

Creating Components

- Schematic cells used in this design:
 1. Qubit: `Q_TransmonCross` / `Q_TransmonPocket` / ...
 2. Resonator / feed: `Q_CpwMeander`, `Q_CpwLine`, `Q_CpwCouplerT`, ...
 3. Ports / pads: `Q_CpwLaunchpadWirebond`, ...
 4. JJ model: `Q_InductorAbstract` (L_J , optional C_J)
 5. Other: interdigital caps, bumps, ...
- Key geometry parameters set in schematic (orientation, lengths, gaps, claws)
- Airbridges: where placed and why
- Quarter-wave sizing checklist:
 1. $\ell = \frac{c}{4f\sqrt{\varepsilon_{\text{eff}}}}$
 2. Record target f , ε_{eff} , resulting ℓ
- Layout screenshots (schematic $\rightarrow$ generated layout)

Electromagnetic Simulation

- Why replace JJ with small-signal L_J in linear EM
- Classical (high power / L_J off) vs dressed (quantum / L_J on) spectra

Simulation Setup

- Ports: launchpad I/O; lumped port at each junction
- Frequency span and local dense sweeps
- Solver choice for this walkthrough (MoM QS / MoM Microwave / FEM)
- Mesh strategy (global cpw; local refine on islands / junctions)
- Materials confirmation checklist
- Figure: port placement / mesh preview

Full EM and S-Parameters

- What you expect in S_{21} / S_{11}

- Peak identification: resonators vs qubits
- Classical vs dressed shift $\rightarrow$ dispersive information
- Troubleshooting if peaks missing / overlapping:
 1. grounds / airbridges
 2. mesh
 3. mode collision
- Figures: S-parameter plots (annotate modes)

Energy Participation Analysis (optional / recommended)

- When you use FEM eigenmode + EPR vs Full EM only
- Mesh notes for EPR
- What you extract: ω_m , Q , T_1 , participations, χ matrix
- Figures: mode fields / participation summary

Quantum Parameter Extraction

- Extract / report:
 1. C (or E_C), anharmonicity α
 2. Detuning Δ , coupling g , cross-Kerr χ
- Compare solvers if you ran more than one (QS / FD / EPR)
- Tuning knobs you actually used:
 1. $L_J \rightarrow$ qubit frequency
 2. resonator length $\rightarrow f_r$
 3. coupler / claw / overlap $\rightarrow g, \chi$
- Results table (fill with your numbers)

Worked Example: [Device Name]

Replace this section title with your actual design (e.g. single Xmon + readout, 5-qubit linear array, flip-chip qubit-resonator).

Technology and Geometry

- Stackup details (single- or multi-layer; thicknesses; air gap if flip-chip)
- Bill of materials / cells and layer prefixes
- Ground stitching / bumps / airbridges
- Figures: full layout (top view; Z-exaggerated if multilayer)

Simulation Results

- Mode frequencies (target vs simulated)
- Extracted α , χ , g , C_{qubit} , ...
- MoM vs FEM comparison (if both)
- Design issues encountered and fixes
- Results table + key plots

Design Notes / Warnings

- Purcell / readout tradeoffs

- Crosstalk, slotline modes, image currents (flip-chip)
- What you would change next iteration

GDS Export and Next Steps

- Export checklist (placeholder inductor layer, GDS options)
- What goes to fab vs what stays simulation-only
- Planned cryogenic measurements to close the loop

Conclusion

- What you designed and simulated
- Main quantitative outcomes
- Lessons for the next chip revision

References

1. TODO: papers and Keysight app notes you relied on
2. P. Krantz et al., “A quantum engineer’s guide to superconducting qubits,” *Appl. Phys. Rev.* **6**, 021318 (2019).
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5. Keysight Technologies, “Design of Transmon Qubit and CPW Resonator in Flip-Chip Technology with ADS, Quantum Pro,” Appl. Note 3125-1427.EN (2025).